

RAJDEEP MUDIAR

B.Tech Computer Science Engineering '28 | Guwahati, Assam, India
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PROFESSIONAL SUMMARY

Computer Science Engineering undergraduate with hands-on experience in **AI/ML, Generative AI, Retrieval-Augmented Generation (RAG), Full-Stack Development, Computer Vision, NLP, and Quantum Communication**. Experienced in building AI-powered applications using **Python, JavaScript, React.js, FastAPI, Node.js, Express.js, LangChain, FAISS, LLMs, and APIs**. Research and internship experience includes RAG-based document question answering, LLM orchestration, quantum key distribution analysis, AI-driven interview evaluation, and full-stack application development. Interested in building practical, scalable AI systems and intelligent applications.

TECHNICAL SKILLS

Programming: Python, JavaScript, Java, HTML, CSS

AI/ML: Artificial Intelligence, Machine Learning, Generative AI, Natural Language Processing, Computer Vision, Sentiment Analysis

LLM / RAG: RAG, LangChain, FAISS, Multi-Query Retrieval, Cross-Encoder Reranking, Conversational Memory, LLM Orchestration, Semantic Search, Ollama, Groq, LangSmith

Full Stack: React.js, Node.js, Express.js, FastAPI, Streamlit, REST APIs, Responsive Web Applications

Databases / Tools: MongoDB, Git, GitHub, API Integration, Authentication, Data Processing

Research: Quantum Computing, Quantum Key Distribution (QKD), Research Methodology, Literature Review, Benchmarking, Performance Analysis

EXPERIENCE

Dev Weekends

June 2026 – September 2026

Full Stack AI Engineering Fellow

- Developed practical full-stack applications using **HTML, CSS, JavaScript, React.js, Node.js, and Express.js**.
- Built responsive and interactive web applications while strengthening frontend and backend development skills.
- Applied debugging, problem-solving, API integration, and modern web development practices across multiple projects.
- Completed hands-on assignments covering frontend development, backend development, and full-stack application architecture.

GitHub: github.com/Rajdeep-Mudiar/dev-fellowship-assignments

Website: dev-fellowship-assignments.vercel.app

Indian Institute of Technology Guwahati

June 2026 – August 2026

Project Intern

- Developed a research-oriented **Retrieval-Augmented Generation (RAG) chatbot** for information retrieval from technical documents and research papers under the guidance of Prof. Prithwijit Guha.
- Implemented document-based conversational question answering using **Streamlit, LangChain, and FAISS**.
- Integrated **Multi-Query Retrieval** and **TinyBERT Cross-Encoder reranking** to improve retrieval relevance.
- Implemented conversational memory and **context-aware question rewriting** for improved multi-turn interactions.
- Built research-focused guardrails and **LaTeX mathematical formula rendering** for technical responses.
- Implemented graph generation from numerical data extracted from technical documents.
- Integrated **Groq-powered LLMs** with **Ollama as a local fallback** and used **LangSmith** for tracing, monitoring, and debugging LLM workflows.

Coding Blocks School of Technology

June 2026 – July 2026

AI Engineer Intern

- Completed a Generative AI-focused internship with practical exposure to **AI-driven applications, machine learning, and NLP**.
- Worked on understanding and implementing AI application workflows and concepts related to Generative AI and ML models.

GitHub: github.com/Rajdeep-Mudiar/CodingBlocks-Assignments

- Conducted research on **Quantum Key Distribution (QKD)** and quantum communication performance across different transmission channels.
- Performed comparative benchmarking and performance analysis across quantum communication channels.
- Extended analysis beyond traditional Fiber and Free-Space Optical (FSO) channels to include **Satellite and Underwater communication channels**.
- Contributed through literature review, technical analysis, research-gap identification, and collaborative discussions.

- Developed an **AI-based interview proctoring and evaluation system** involving real-time monitoring, speech processing, and intelligent response analysis.
- Implemented backend functionality using **Python**, including API development, authentication handling, and data-processing workflows.
- Built features for **speech-to-text conversion, candidate response evaluation, and session management**.
- Collaborated on responsive frontend functionality, resolved dependency conflicts, debugged system-level issues, and optimized application performance.

PROJECTS & HACKATHONS

CareBridge – AI-Powered Elderly Healthcare Platform

ET AI Hackathon 2026 | Semi-Finalist

- Proposed an AI, IoT, Ambient Intelligence, and Digital Twin-based platform for proactive elderly healthcare.
- Designed a **zero-UI passive monitoring** approach with AI-driven anomaly detection and environment-plus-vitals-based risk intelligence.
- Proposed voice-first multilingual assistance, rural-ready **BLE + LoRa architecture**, Digital Twin visualization, and environmental factor integration.
- Selected as a **Semi-Finalist among 55,000+ teams** in the ET AI Hackathon 2026.

SahayaKISSAN – Smart Agriculture Assistance Platform

Gauhati University

- Contributed to an **AI-powered agriculture platform** for crop monitoring and farmer assistance.
- Worked on **computer vision and AI modules** for crop health analysis and detection.
- Collaborated using GitHub for version control and assisted in integrating AI outputs with farmer-facing components.

GitHub: github.com/Rajdeep-Mudiar/SahayaKISSAN

HackDays 3.0 – Team CodeBusters

Gauhati University | March 2025

- Built an AI-powered platform addressing **education, mental health, and disaster management**.
- Developed chatbots, a therapy platform, and real-time alert systems using **Node.js, Express.js, MongoDB, and APIs**.
- Integrated external APIs including **OpenWeather and GDACS** and gained experience in full-stack development and API integration.

EDUCATION
